The LibertinusT1 Math Package

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This package is a LaTeX math package to accompany the LaTeX Libertine text package, serving as an alternative to `newtxmath` with the `libertine` option. Both packages use the Greek and Roman letters from Libertine, but LibertinusT1Math symbols are mostly taken from the LibertinusMath font which extends and corrects Libertine, while `newtxmath` uses symbols matching Times in weight and size, and so are a bit heavier and larger. I've changed a number of glyphs from LibertinusMath and added dozens of new glyphs. Like LibertinusMath, LibertinusT1Math does not contain calligraphic, script or gothic alphabets. The package sets `\mathcal` to the (unscaled) STIX calligraphic font and `\mathbb` to the LibertinusT1Math double-struck alphabet. The `mathalpha` package (loaded AFTER `libertinust1math`) offers a convenient way to change these default choices and to set the script and gothic alphabets. See the sample invocation below, which assumes you have `mathalpha` version 1.08 (March 2016) or higher.

Additions in version 1.1: There is now support for sans serif fonts in math that allows the ISO typesetting rules to be followed, and to provide a math processing system using sans serif Roman and Greek letters. See the end of this documentation for details.

Additions in version 1.1.8: There is now support for the scaling options in both the old libertine and the new libertinus[-type1] packages. Prior versions supported only earlier libertine package[s]. The details are spelled out below in the discussion of the `scaled` package option.

Changes and Additions in version 2.0: Prior versions of `libertinust1math` were based on the glyphs in `LibertinusMath.otf` from late 2015. This version is based on the late 2021 font which made a number of important changes.

- The math axis changed from 251 to 260.
- The symbol sizes were enlarged substantially.

The overly small symbols of prior versions are no longer a concern. Additions include some of the features of `newtxmath`: a `subscriptcorrection` option to improve the kerning of some problem glyphs such as j, f,β in subscripts; an adaptive vector accent with macro `\vv` so that you can write `$\vv{XY}$` to get $\overrightarrow{XY}$, placing an arrow of appropriate length over any mathematical expression; a `\widebar` macro was copied from `newtxmath` so that `$\widebar{XY}$` results in $\overline{XY}$. See the documentation of `newtx` for further details on these.

Package Options:

- **New in v.2.0** `subscriptcorrection` allows for a number of problem letters to behave better in subscripts, but at the cost of compatibility with the `xy` package.
- **New in v.2.0** `amsthm` loads the corresponding package in the proper order relative to `amsmath`. `lcgreekalpha` makes lower-case Greek letters be of type `mathalpha` rather than `mathord`, forcing them to respond to alphabet changing commands such as `\mathnormal`, `\mathrm`, `\mathit` and `\mathsf`.
- `upint` changes the style of integrals from the default *slanted* $\int$ to *upright* $\int$.
- `frenchmath` changes the math style to use upright Greek and Roman capital letters rather than math italic shapes.
- `slantedGreek` makes uppercase Greek math letters slanted.
- `uprightGreek` makes uppercase Greek math letters upright. This is the default except in ISO mode.

- scaled (or scale may be used change the scale of the math. The fd files in this package respond to three scale-changing macros:
  - \LinuxLibertineT@scale, set from option llscale (or llscaled) to the libertine package;
  - \LibertinusRM@scale, set by option ScaleRM to the libertinus-type1 package or to the libertinus package;
  - \libus@scale, set by the scaled option to libertinust1math.

  The current libertineRoman package contains no option that affects the scale parameter. If you need to scale libertineRoman, add to your preamble lines like

  ```
  \makeatletter
  \newcommand*{\LinuxLibertineT@scale}{1.03}
  \makeatother
  ```

  If \libus@scale is defined, it takes precedence over the other two settings. The result on math scaling only of the different possible scale settings is:
  - the scaled option to libertinust1math overrides all other scale settings;
  - if you did not issue a scale or scaled option to libertinust1math, then
    * if you used a libertine package that set a value for \LinuxLibertineT@scale, then its value is used;
    * otherwise, if you used libertinus-type1 or libertinus, then the value of \LibertinusRM@scale is used.
- See the new material at the end for a discussion of the new options connected with sans serif in math.

Sample Invocation:

```
\usepackage[sb]{libertine} % or \usepackage[sb]{libertinus}
\usepackage[T1]{fontenc}
\usepackage{textcomp}
\usepackage[varqu,varl]{zi4}% inconsolata for mono, not LibertineMono
\usepackage[amsthm]{libertinust1math} % slanted integrals, by default
\usepackage[scr=boondoxo,bb=boondox]{mathalpha} %Omit bb=boondox for default libertinus bb
```

Some Things to Note:

- The amsmath package is loaded, if necessary, by libertinust1math. The package depends on a number of its features and improvements to math typesetting.
- There are no bold symbols in LibertinusMath, but there are bold Roman and Greek letters, and the same is true of LibertinusT1Math.
- The italic letter v (v) in LibertinusMath is easily confused with Greek \nu (ν), so the math italic letter v is set to υ, and similarly in the bold math version.
- The layout of the type1 math fonts follows STIX to some extent, as does the sty file, though the content of this package is much less rich.
- Dotlessi and dotlessj are available in four math styles—mathit, mathrm, mathbf and mathbfit. For example:

Macro	Result
$\imath$	$\imath$
$\mathrm{\imath}$	$\mathrm{\imath}$
$\mathbf{\imath}$	$\mathbf{\imath}$
$\mathbfit{\imath}$	$\mathbfit{\imath}$

- There are Greek alphabets in upright and italic shapes, regular and bold weights. Upright Greek letters have names like `\alphaup` (and `\upalpha`) while the italic forms have names like `\alphait`. According to the options you set, `\alpha` and so on are `\let` to the appropriate choice, but you may always use the underlying forms if you wish to use both upright and italic forms in your document.
- Uppercase Greek letters are of type `mathalpha`, so when you write `$\mathit{\Gamma}$`, you'll get $\mathit{\Gamma}$. However, unless you chose the option `lcgreekalpha`, `$\mathrm{\beta}$` would produce β, as it is of type `mathord` by default.
- There is an upright partial derivative symbol named `\uppartial` that typesets as ∂, for those who wish to follow ISO rules.
- There are (as of version 1.1) macros for mathematical sans serif: `\mathsf`, `\mathsfit` (italic sans), `\mathsfbf` (bold sans) and `\mathsfbfit` (bold sans italic.) See the end of this documentation for details.
- If you use one of the AMS classes such as `amsart` or `amsbook`, then the AMS fonts will be loaded even though not needed, taking up two of your precious sixteen math families. If you are running out of families ("too many math alphabets used in version normal"), use another capable class such as `memoir` or `scrartcl` and gain two more families.

The next pages show the font tables for LibertinusT1Math. If you see an unfamiliar symbol and wish to learn its LaTeX name, get the LaTeX name of the font (e.g., `operators`, `letters`, `symbols`, `largesymbols`) and the hex location of the glyph (e.g., `"FF` is hex notation for 255) and search `libertinust1math.sty` by hex number.

libertinust1-mathrm (operators):

	´0	´1	´2	´3	´4	´5	´6	´7	
´00x	Γ	Δ	Θ	Λ	Ξ	Π	Σ	Υ	″0x
´01x	Φ	Ψ	Ω	α	β	γ	δ	ϵ	
´02x	ζ	η	θ	ι	κ	λ	μ	ν	″1x
´03x	ξ	π	ρ	σ	τ	υ	ϕ	χ	
´04x	ψ	ω	ε	ϑ	ϖ	ϱ	ς	φ	″2x
´05x	∇	∂	−	+	±	∓	(	)	
´06x	0	1	2	3	4	5	6	7	″3x
´07x	8	9	:	;	*	=	$	?	
´10x	!	A	B	C	D	E	F	G	″4x
´11x	H	I	J	K	L	M	N	O	
´12x	P	Q	R	S	T	U	V	W	″5x
´13x	X	Y	Z	[	\	]	{	/	
´14x	}	a	b	c	d	e	f	g	″6x
´15x	h	i	j	k	l	m	n	o	
´16x	p	q	r	s	t	u	v	w	″7x
´17x	x	y	z	ı	ȷ	#	%	’	
´20x	̀	́	̂	̃	̄	̆	̇	̈	″8x
´21x	̊	̋	̌	̑	̒	̓	̛	̲	
´22x	⃗	⃖	→	...	....	↔	⌐	¬	″9x
´23x	*	&	@	¬	·	×	⪯	÷	
´24x	Ƶ	⅊	ɜ	†	‡	•	..	…	″Ax
´25x	/	//	///	\	\\	\\\	‸	‼	
´26x	▪	⁄	??		////	∤			″Bx
´27x	≍		Ɛ	℧	ɪ	Å	Ǝ	Ↄ	
´30x	⅂	⊥	ʎ		⅋	∀	Ɔ	∃	″Cx
´31x	∄	∅	Δ	∈	∉	ϵ	∋	∌	
´32x	ɜ	∎	∔	≥	⟍	∘	•	∝	″Dx
´33x	∞	⌞	∠	∡	∢	∣	∤	∥	
´34x	∦	∧	∨	∩	∪	∴	∵	∅	″Ex
´35x	::	∸	∹	∺	÷	∼	∽	∾	
´36x	∿	≀	≁	≃	≄	≅	≇	≉	″Fx
´37x	≉	≈	≉	≊	≋	≌	ϰ	≎	
	″8	″9	″A	″B	″C	″D	″E	″F	

libertinust1-mathrm (bold-operators):

	´0	´1	´2	´3	´4	´5	´6	´7	
´00x	$\mathbf{\Gamma}_{0}$	$\mathbf{\Delta}_{1}$	$\mathbf{\Theta}_{2}$	$\mathbf{\Lambda}_{3}$	$\mathbf{\Xi}_{4}$	$\mathbf{\Pi}_{5}$	$\mathbf{\Sigma}_{6}$	$\mathbf{\Upsilon}_{7}$	˝0x
´01x	$\mathbf{\Phi}_{8}$	$\mathbf{\Psi}_{9}$	$\mathbf{\Omega}_{10}$	$\boldsymbol{\alpha}_{11}$	$\boldsymbol{\beta}_{12}$	$\boldsymbol{\gamma}_{13}$	$\boldsymbol{\delta}_{14}$	$\boldsymbol{\epsilon}_{15}$	
´02x	$\boldsymbol{\zeta}_{16}$	$\boldsymbol{\eta}_{17}$	$\boldsymbol{\theta}_{18}$	$\boldsymbol{\iota}_{19}$	$\boldsymbol{\kappa}_{20}$	$\boldsymbol{\lambda}_{21}$	$\boldsymbol{\mu}_{22}$	$\boldsymbol{\nu}_{23}$	˝1x
´03x	$\boldsymbol{\xi}_{24}$	$\boldsymbol{\pi}_{25}$	$\boldsymbol{\rho}_{26}$	$\boldsymbol{\sigma}_{27}$	$\boldsymbol{\tau}_{28}$	$\boldsymbol{\upsilon}_{29}$	$\boldsymbol{\phi}_{30}$	$\boldsymbol{\chi}_{31}$	
´04x	$\boldsymbol{\psi}_{32}$	$\boldsymbol{\omega}_{33}$	$\boldsymbol{\varepsilon}_{34}$	$\boldsymbol{\vartheta}_{35}$	$\boldsymbol{\varpi}_{36}$	$\boldsymbol{\varrho}_{37}$	$\boldsymbol{\varsigma}_{38}$	$\boldsymbol{\varphi}_{39}$	˝2x
´05x	$\boldsymbol{\nabla}_{40}$	$\boldsymbol{\partial}_{41}$	$-_{42}$	$+_{43}$	$\pm_{44}$	$\mp_{45}$	$(_{46}$	$)_{47}$	
´06x	$\mathbf{0}_{48}$	$\mathbf{1}_{49}$	$\mathbf{2}_{50}$	$\mathbf{3}_{51}$	$\mathbf{4}_{52}$	$\mathbf{5}_{53}$	$\mathbf{6}_{54}$	$\mathbf{7}_{55}$	˝3x
´07x	$\mathbf{8}_{56}$	$\mathbf{9}_{57}$	$:_{58}$	$;_{59}$	$*_{60}$	$=_{61}$	$\$_{62}$	$?_{63}$	
´10x	$!_{64}$	$\mathbf{A}_{65}$	$\mathbf{B}_{66}$	$\mathbf{C}_{67}$	$\mathbf{D}_{68}$	$\mathbf{E}_{69}$	$\mathbf{F}_{70}$	$\mathbf{G}_{71}$	˝4x
´11x	$\mathbf{H}_{72}$	$\mathbf{I}_{73}$	$\mathbf{J}_{74}$	$\mathbf{K}_{75}$	$\mathbf{L}_{76}$	$\mathbf{M}_{77}$	$\mathbf{N}_{78}$	$\mathbf{O}_{79}$	
´12x	$\mathbf{P}_{80}$	$\mathbf{Q}_{81}$	$\mathbf{R}_{82}$	$\mathbf{S}_{83}$	$\mathbf{T}_{84}$	$\mathbf{U}_{85}$	$\mathbf{V}_{86}$	$\mathbf{W}_{87}$	˝5x
´13x	$\mathbf{X}_{88}$	$\mathbf{Y}_{89}$	$\mathbf{Z}_{90}$	$[_{91}$	$\backslash_{92}$	$]_{93}$	$\{_{94}$	$/_{95}$	
´14x	$\}_{96}$	$\mathbf{a}_{97}$	$\mathbf{b}_{98}$	$\mathbf{c}_{99}$	$\mathbf{d}_{100}$	$\mathbf{e}_{101}$	$\mathbf{f}_{102}$	$\mathbf{g}_{103}$	˝6x
´15x	$\mathbf{h}_{104}$	$\mathbf{i}_{105}$	$\mathbf{j}_{106}$	$\mathbf{k}_{107}$	$\mathbf{l}_{108}$	$\mathbf{m}_{109}$	$\mathbf{n}_{110}$	$\mathbf{o}_{111}$	
´16x	$\mathbf{p}_{112}$	$\mathbf{q}_{113}$	$\mathbf{r}_{114}$	$\mathbf{s}_{115}$	$\mathbf{t}_{116}$	$\mathbf{u}_{117}$	$\mathbf{v}_{118}$	$\mathbf{w}_{119}$	˝7x
´17x	$\mathbf{x}_{120}$	$\mathbf{y}_{121}$	$\mathbf{z}_{122}$	$\mathbf{1}_{123}$	$\mathbf{J}_{124}$	$\#_{125}$	$\%_{126}$	$'_{127}$	
´20x	$\grave{}_{128}$	$\acute{}_{129}$	$\hat{}_{130}$	$\tilde{}_{131}$	$\bar{}_{132}$	$\breve{}_{133}$	$\dot{}_{134}$	$\ddot{}_{135}$	˝8x
´21x	$\mathring{}_{136}$	$\circ_{137}$	$\check{}_{138}$	$\mathring{}_{139}$	$\grave{}_{140}$	$\acute{}_{141}$	$\urcorner_{142}$	$\leftarrow_{143}$	
´22x	$\rightarrow_{144}$	$\leftarrow_{145}$	$\rightarrow_{146}$	$\cdots_{147}$	$\cdots_{148}$	$\leftrightarrow_{149}$	$\top_{150}$	$\neg_{151}$	˝9x
´23x	$*_{152}$	$\&_{153}$	$@_{154}$	$\neg_{155}$	$\cdot_{156}$	$\times_{157}$	$\preceq_{158}$	$\div_{159}$	
´24x	$\mathbb{Z}_{160}$	$\not{}_{161}$	$\ni_{162}$	$\dagger_{163}$	$\ddagger_{164}$	$\bullet_{165}$	$..._{166}$	$\ldots_{167}$	˝Ax
´25x	$/_{168}$	$//_{169}$	$///_{170}$	$\backslash_{171}$	$\backslash\backslash_{172}$	$\backslash\backslash\backslash_{173}$	$_{\wedge 174}$	$!!_{175}$	
´26x	$\blacksquare_{176}$	$\diagup_{177}$	$??_{178}$	$_{179}$	$////_{180}$	$\nmid_{181}$	$_{182}$	$_{183}$	˝Bx
´27x	$\asymp_{184}$	$_{185}$	$\mathfrak{E}_{186}$	$\mathfrak{G}_{187}$	$\imath_{188}$	$\mathring{A}_{189}$	$\exists_{190}$	$\mathfrak{O}_{191}$	
´30x	$\urcorner_{192}$	$\bot_{193}$	λ_{194}	$_{195}$	$\wp_{196}$	$\forall_{197}$	$\complement_{198}$	$\exists_{199}$	˝Cx
´31x	$\nexists_{200}$	$\emptyset_{201}$	Δ_{202}	$\in_{203}$	$\notin_{204}$	$\in_{205}$	$\ni_{206}$	$\not\ni_{207}$	
´32x	$\ni_{208}$	$\blacksquare_{209}$	$\dot{+}_{210}$	$\succeq_{211}$	$\backslash_{212}$	$\circ_{213}$	$\bullet_{214}$	$\propto_{215}$	˝Dx
´33x	∞_{216}	$\llcorner_{217}$	$\angle_{218}$	$\measuredangle_{219}$	$\sphericalangle_{220}$	$\mid_{221}$	$\nmid_{222}$	$\parallel_{223}$	
´34x	$\nparallel_{224}$	$\wedge_{225}$	$\vee_{226}$	$\cap_{227}$	$\cup_{228}$	$\therefore_{229}$	$\because_{230}$	$\emptyset_{231}$	˝Ex
´35x	$::_{232}$	$\dot{-}_{233}$	$-\!\!:_{234}$	$\div_{235}$	$\div_{236}$	$\sim_{237}$	$\backsim_{238}$	$\wr_{239}$	
´36x	$\backsim_{240}$	$\backsim_{241}$	$\backsim_{242}$	$\simeq_{243}$	$\simeq_{244}$	$\not\simeq_{245}$	$\cong_{246}$	$\not\cong_{247}$	˝Fx
´37x	$\not\cong_{248}$	$\approx_{249}$	$\not\approx_{250}$	$\approx\!\!=_{251}$	$\approx_{252}$	$\eqsim_{253}$	$\boldsymbol{\varkappa}_{254}$	$\fallingdotseq_{255}$	
	˝8	˝9	˝A	˝B	˝C	˝D	˝E	˝F	

libertinust1-mathit (letters):

	´0	´1	´2	´3	´4	´5	´6	´7	
´00x	Γ_{0}	Δ_{1}	Θ_{2}	Λ_{3}	Ξ_{4}	Π_{5}	Σ_{6}	Y_{7}	″0x
´01x	Φ_{8}	Ψ_{9}	Ω_{10}	α_{11}	β_{12}	γ_{13}	δ_{14}	ϵ_{15}	
´02x	ζ_{16}	η_{17}	θ_{18}	ι_{19}	κ_{20}	λ_{21}	μ_{22}	ν_{23}	″1x
´03x	ξ_{24}	π_{25}	ρ_{26}	σ_{27}	τ_{28}	υ_{29}	ϕ_{30}	χ_{31}	
´04x	ψ_{32}	ω_{33}	ε_{34}	ϑ_{35}	ϖ_{36}	ϱ_{37}	ς_{38}	φ_{39}	″2x
´05x	∇_{40}	∂_{41}	$\aleph_{42}$	$\beth_{43}$	$\gimel_{44}$	$\daleth_{45}$	$\triangleright_{46}$	$\triangleleft_{47}$	
´06x	0_{48}	1_{49}	2_{50}	3_{51}	4_{52}	5_{53}	6_{54}	7_{55}	″3x
´07x	8_{56}	9_{57}	$.{}_{58}$	$,{}_{59}$	$<_{60}$	$\hbar_{61}$	$>_{62}$	$\star_{63}$	
´10x	$\lneqq_{64}$	A_{65}	B_{66}	C_{67}	D_{68}	E_{69}	F_{70}	G_{71}	″4x
´11x	H_{72}	I_{73}	J_{74}	K_{75}	L_{76}	M_{77}	N_{78}	O_{79}	
´12x	P_{80}	Q_{81}	R_{82}	S_{83}	T_{84}	U_{85}	V_{86}	W_{87}	″5x
´13x	X_{88}	Y_{89}	Z_{90}	$\flat_{91}$	$\natural_{92}$	$\sharp_{93}$	$\smile_{94}$	$\frown_{95}$	
´14x	$\hbar_{96}$	a_{97}	b_{98}	c_{99}	d_{100}	e_{101}	f_{102}	g_{103}	″6x
´15x	h_{104}	i_{105}	j_{106}	k_{107}	l_{108}	m_{109}	n_{110}	o_{111}	
´16x	p_{112}	q_{113}	r_{114}	s_{115}	t_{116}	u_{117}	v_{118}	w_{119}	″7x
´17x	x_{120}	y_{121}	z_{122}	$\imath_{123}$	$\jmath_{124}$	$\gneqq_{125}$	$\ll_{126}$	$\frown_{127}$	
´20x	$\grave{}_{128}$	$\acute{}_{129}$	$\hat{}_{130}$	$\tilde{}_{131}$	$\bar{}_{132}$	$\breve{}_{133}$	$\dot{}_{134}$	$\ddot{}_{135}$	″8x
´21x	$\mathring{}_{136}$	$\mathring{}_{137}$	$\check{}_{138}$	$\check{}_{139}$	$\grave{}_{140}$	$\acute{}_{141}$	$\dot{}_{142}$	$\ddot{}_{143}$	
´22x	$\vec{}_{144}$	$\overleftarrow{}_{145}$	$\vec{}_{146}$	$\dddot{}_{147}$	$\ddddot{}_{148}$	$\overleftrightarrow{}_{149}$	$\urcorner_{150}$	$\urcorner_{151}$	″9x
´23x	$*_{152}$	$\bar{}_{153}$	$\hat{}_{154}$	$\tilde{}_{155}$	$\check{}_{156}$	$\hat{}_{157}$	$\tilde{}_{158}$	$\check{}_{159}$	
´24x	$\overbrace{}^{160}$	$\overbrace{}^{161}$	$\overbrace{}^{162}$	$\overbrace{}^{163}$	$\overbrace{}^{164}$	$\overbrace{}^{165}$	$\overbrace{}^{166}$	$\overbrace{}^{167}$	″Ax
´25x	$\underbrace{}_{168}$	$\llcorner_{169}$	$\lnot_{170}$	$\backsim_{171}$	$\backsim_{172}$	$\multimap_{173}$	$\rightsquigarrow_{174}$	$\rightsquigarrow_{175}$	
´26x	$\overparen{}_{176}$	$\overparen{}_{177}$	$\underparen{}_{178}$	$\underparen{}_{179}$	$\ulcorner_{180}$	$\urcorner_{181}$	$\llcorner_{182}$	$\lrcorner_{183}$	″Bx
´27x	$\gg_{184}$	$\ll_{185}$	$\precneqq_{186}$	$\nprec_{187}$	$\succneqq_{188}$	$\nleq_{189}$	$\ngeq_{190}$	$\lesssim_{191}$	
´30x	$\gtrsim_{192}$	$\nleqslant_{193}$	$\ngeqslant_{194}$	$\lesseqgtr_{195}$	$\gtreqless_{196}$	$\nleqslant_{197}$	$\ngeqslant_{198}$	$\prec_{199}$	″Cx
´31x	$\succ_{200}$	$\preccurlyeq_{201}$	$\succcurlyeq_{202}$	$\precsim_{203}$	$\succsim_{204}$	$\nprec_{205}$	$\nsucc_{206}$	$\subset_{207}$	
´32x	$\supset_{208}$	$\not\subset_{209}$	$\not\supset_{210}$	$\subseteq_{211}$	$\supseteq_{212}$	$\not\subseteq_{213}$	$\not\supseteq_{214}$	$\subsetneq_{215}$	″Dx
´33x	$\supsetneq_{216}$	$\uplus_{217}$	$\uplus_{218}$	$\uplus_{219}$	$\sqsubset_{220}$	$\sqsupset_{221}$	$\sqsubseteq_{222}$	$\sqsupseteq_{223}$	
´34x	$\sqcap_{224}$	$\sqcup_{225}$	$\oplus_{226}$	$\ominus_{227}$	$\otimes_{228}$	$\oslash_{229}$	$\odot_{230}$	$\circledcirc_{231}$	″Ex
´35x	$\circledast_{232}$	$\circleddash_{233}$	$\circleddash_{234}$	$\boxplus_{235}$	$\boxminus_{236}$	$\boxtimes_{237}$	$\boxdot_{238}$	$\vdash_{239}$	
´36x	$\dashv_{240}$	$\top_{241}$	$\bot_{242}$	$\vdash_{243}$	$\models_{244}$	$\models_{245}$	$\Vdash_{246}$	$\Vvdash_{247}$	″Fx
´37x	$\models_{248}$	$\nvdash_{249}$	$\nvDash_{250}$	$\nVdash_{251}$	$\nVDash_{252}$	$_{253}$	$\varkappa_{254}$	$\trianglelefteq_{255}$	
	″8	″9	″A	″B	″C	″D	″E	″F	

libertinust1-mathit-bold (bold-letters):

	´0	´1	´2	´3	´4	´5	´6	´7	
´00x	$\boldsymbol{\Gamma}_{0}$	$\boldsymbol{\Delta}_{1}$	$\boldsymbol{\Theta}_{2}$	$\boldsymbol{\Lambda}_{3}$	$\boldsymbol{\Xi}_{4}$	$\boldsymbol{\Pi}_{5}$	$\boldsymbol{\Sigma}_{6}$	$\boldsymbol{Y}_{7}$	″0x
´01x	$\boldsymbol{\Phi}_{8}$	$\boldsymbol{\Psi}_{9}$	$\boldsymbol{\Omega}_{10}$	$\boldsymbol{\alpha}_{11}$	$\boldsymbol{\beta}_{12}$	$\boldsymbol{\gamma}_{13}$	$\boldsymbol{\delta}_{14}$	$\boldsymbol{\epsilon}_{15}$	
´02x	$\boldsymbol{\zeta}_{16}$	$\boldsymbol{\eta}_{17}$	$\boldsymbol{\theta}_{18}$	$\boldsymbol{\imath}_{19}$	$\boldsymbol{\kappa}_{20}$	$\boldsymbol{\lambda}_{21}$	$\boldsymbol{\mu}_{22}$	$\boldsymbol{\nu}_{23}$	″1x
´03x	$\boldsymbol{\xi}_{24}$	$\boldsymbol{\pi}_{25}$	$\boldsymbol{\rho}_{26}$	$\boldsymbol{\sigma}_{27}$	$\boldsymbol{\tau}_{28}$	$\boldsymbol{\upsilon}_{29}$	$\boldsymbol{\phi}_{30}$	$\boldsymbol{\chi}_{31}$	
´04x	$\boldsymbol{\psi}_{32}$	$\boldsymbol{\omega}_{33}$	$\boldsymbol{\varepsilon}_{34}$	$\boldsymbol{\vartheta}_{35}$	$\boldsymbol{\varpi}_{36}$	$\boldsymbol{\varrho}_{37}$	$\boldsymbol{\varsigma}_{38}$	$\boldsymbol{\varphi}_{39}$	″2x
´05x	$\boldsymbol{\nabla}_{40}$	$\boldsymbol{\partial}_{41}$	$\aleph_{42}$	$\beth_{43}$	$\gimel_{44}$	$\daleth_{45}$	$\triangleright_{46}$	$\triangleleft_{47}$	
´06x	$\mathbf{0}_{48}$	$\mathbf{1}_{49}$	$\mathbf{2}_{50}$	$\mathbf{3}_{51}$	$\mathbf{4}_{52}$	$\mathbf{5}_{53}$	$\mathbf{6}_{54}$	7_{55}	″3x
´07x	$\mathbf{8}_{56}$	$\mathbf{9}_{57}$	.58	,59	$<_{60}$	$\hbar_{61}$	$>_{62}$	$\star_{63}$	
´10x	$\lneqq_{64}$	$\boldsymbol{A}_{65}$	$\boldsymbol{B}_{66}$	$\boldsymbol{C}_{67}$	$\boldsymbol{D}_{68}$	$\boldsymbol{E}_{69}$	$\boldsymbol{F}_{70}$	$\boldsymbol{G}_{71}$	″4x
´11x	$\boldsymbol{H}_{72}$	$\boldsymbol{I}_{73}$	$\boldsymbol{J}_{74}$	$\boldsymbol{K}_{75}$	$\boldsymbol{L}_{76}$	$\boldsymbol{M}_{77}$	$\boldsymbol{N}_{78}$	$\boldsymbol{O}_{79}$	
´12x	$\boldsymbol{P}_{80}$	$\boldsymbol{Q}_{81}$	$\boldsymbol{R}_{82}$	$\boldsymbol{S}_{83}$	$\boldsymbol{T}_{84}$	$\boldsymbol{U}_{85}$	$\boldsymbol{V}_{86}$	$\boldsymbol{W}_{87}$	″5x
´13x	$\boldsymbol{X}_{88}$	$\boldsymbol{Y}_{89}$	$\boldsymbol{Z}_{90}$	$\flat_{91}$	$\natural_{92}$	$\sharp_{93}$	$\smile_{94}$	$\frown_{95}$	
´14x	$\hbar_{96}$	$\boldsymbol{a}_{97}$	$\boldsymbol{b}_{98}$	$\boldsymbol{c}_{99}$	$\boldsymbol{d}_{100}$	$\boldsymbol{e}_{101}$	$\boldsymbol{f}_{102}$	$\boldsymbol{g}_{103}$	″6x
´15x	$\boldsymbol{h}_{104}$	$\boldsymbol{i}_{105}$	$\boldsymbol{j}_{106}$	$\boldsymbol{k}_{107}$	$\boldsymbol{l}_{108}$	$\boldsymbol{m}_{109}$	$\boldsymbol{n}_{110}$	$\boldsymbol{o}_{111}$	
´16x	$\boldsymbol{p}_{112}$	$\boldsymbol{q}_{113}$	$\boldsymbol{r}_{114}$	$\boldsymbol{s}_{115}$	$\boldsymbol{t}_{116}$	$\boldsymbol{u}_{117}$	$\boldsymbol{v}_{118}$	$\boldsymbol{w}_{119}$	″7x
´17x	$\boldsymbol{x}_{120}$	$\boldsymbol{y}_{121}$	$\boldsymbol{z}_{122}$	$\boldsymbol{\imath}_{123}$	$\boldsymbol{\jmath}_{124}$	$\gneqq_{125}$	$\ll_{126}$	$\frown_{127}$	
´20x	$\grave{}_{128}$	$\acute{}_{129}$	$\hat{}_{130}$	$\tilde{}_{131}$	$\bar{}_{132}$	$\breve{}_{133}$	$\dot{}_{134}$	$\ddot{}_{135}$	″8x
´21x	$\mathring{}_{136}$	$\circ_{137}$	$\check{}_{138}$	$\overset{\frown}{}_{139}$	$\grave{}_{140}$	$\acute{}_{141}$	$\urcorner_{142}$	$\overset{\leftarrow}{}_{143}$	
´22x	$\overset{\rightarrow}{}_{144}$	$\overset{\leftarrow}{}_{145}$	$\overset{\rightarrow}{}_{146}$	$\dddot{}_{147}$	$\ddddot{}_{148}$	$\overset{\leftrightarrow}{}_{149}$	$\urcorner_{150}$	$\urcorner_{151}$	″9x
´23x	$*_{152}$	$-_{153}$	$\widehat{}_{154}$	$\sim_{155}$	$\vee_{156}$	$\widehat{}_{157}$	$\sim_{158}$	$\vee_{159}$	
´24x	$\widehat{}_{160}$	$\sim_{161}$	$\vee_{162}$	$\widehat{}_{163}$	$\sim_{164}$	$\vee_{165}$	$\widehat{}_{166}$	$\sim_{167}$	″Ax
´25x	$\vee_{168}$	$\backsim_{169}$	$\neg_{170}$	$\backsim_{171}$	$\backsim_{172}$	$-_{173}$	$\backsim_{174}$	$\backsim_{175}$	
´26x	$\overset{\frown}{}_{176}$	$\overset{\frown}{}_{177}$	$\overset{\smile}{}_{178}$	$\overset{\smile}{}_{179}$	$\ulcorner_{180}$	$\urcorner_{181}$	$\llcorner_{182}$	$\lrcorner_{183}$	″Bx
´27x	$\gg_{184}$	$\between_{185}$	$\neq_{186}$	$\not<_{187}$	$\not>_{188}$	$\not\leqslant_{189}$	$\not\geqslant_{190}$	$\lesssim_{191}$	
´30x	$\gtrsim_{192}$	$\not\lesssim_{193}$	$\not\gtrsim_{194}$	$\lessgtr_{195}$	$\gtrless_{196}$	$\not\lessgtr_{197}$	$\not\gtrless_{198}$	$\prec_{199}$	″Cx
´31x	$\succ_{200}$	$\preccurlyeq_{201}$	$\succcurlyeq_{202}$	$\precsim_{203}$	$\succsim_{204}$	$\not\prec_{205}$	$\not\succ_{206}$	$\subset_{207}$	
´32x	$\supset_{208}$	$\not\subset_{209}$	$\not\supset_{210}$	$\subseteq_{211}$	$\supseteq_{212}$	$\not\subseteq_{213}$	$\not\supseteq_{214}$	$\subsetneq_{215}$	″Dx
´33x	$\supsetneq_{216}$	$\uplus_{217}$	$\uplus_{218}$	$\uplus_{219}$	$\sqsubset_{220}$	$\sqsupset_{221}$	$\sqsubseteq_{222}$	$\sqsupseteq_{223}$	
´34x	$\sqcap_{224}$	$\sqcup_{225}$	$\oplus_{226}$	$\ominus_{227}$	$\otimes_{228}$	$\oslash_{229}$	$\odot_{230}$	$\circledcirc_{231}$	″Ex
´35x	$\circledast_{232}$	$\circleddash_{233}$	$\ominus_{234}$	$\boxplus_{235}$	$\boxminus_{236}$	$\boxtimes_{237}$	$\boxdot_{238}$	$\vdash_{239}$	
´36x	$\dashv_{240}$	$\top_{241}$	$\bot_{242}$	$\vdash_{243}$	$\models_{244}$	$\models_{245}$	$\Vdash_{246}$	$\Vvdash_{247}$	″Fx
´37x	$\Vdash_{248}$	$\nvdash_{249}$	$\nvDash_{250}$	$\nVdash_{251}$	$\nVDash_{252}$	253	$\boldsymbol{\varkappa}_{254}$	$\trianglelefteq_{255}$	
	″8	″9	″A	″B	″C	″D	″E	″F	

libertinust1-mathsym (symbols):

	´0	´1	´2	´3	´4	´5	´6	´7	
´00x	⊵0	⊶1	⊷2	∫3	∬4	∭5	∮6	⨌7	″0x
´01x	∫8	∬9	∭10	∮11	·12	⨌13	⋈14	⋉15	
´02x	⋊16	∏17	⨿18	≔19	⩛20	⩓21	∑22	⊥23	″1x
´03x	ℌ24	⧹25	⧸26	⟙27	⋖28	⋗29	⋘30	⋙31	
´04x	≦32	≧33	≺34	≻35	⪷36	⪸37	⪺38	⪻39	″2x
´05x	⊈40	⊉41	⊏42	⊐43	≲44	≳45	⊀46	⊁47	
´06x	⋪48	⋫49	⋬50	⋭51	⋮52	⋯53	⋰54	⋱55	″3x
´07x	⋩56	⋨57	⋦58	⋧59	⋩60	⋨61	62	⨍63	
´10x	∋64	ᶜ65	ᵓ66	←67	↑68	→69	↓70	↔71	″4x
´11x	↕72	↖73	↗74	↘75	↙76	↜77	↛78	↤79	
´12x	↦80	↩81	↪82	↮83	↼84	↽85	⇀86	⇁87	″5x
´13x	⇋88	⇌89	⟵90	⟶91	⟷92	⟻93	⟼94	∅95	
´14x	≏96	≑97	÷98	≒99	≓100	≔101	≕102	≖103	″6x
´15x	≗104	≘105	≙106	≚107	≛108	≜109	≝110	≞111	
´16x	≟112	≠113	≡114	≢115	≀116	≬117	≣118	≤119	″7x
´17x	≥120	≦121	≧122	ℜ123	ℑ124	℘125	⌓126	⌒127	
´20x	♣128	◆129	♥130	♠131	♪132	°133	§134	¶135	″8x
´21x	✓136	‾137	‾138	ℓ139	⅋140	⅁141	142	143	
´22x	144	145	146	147	148	149	150	151	″9x
´23x	152	153	154	⌜155	156	157	158	159	
´26x	176	177	178	179	180	181	␣182	■183	″Bx
´27x	□184	185	186	187	188	189	190	191	
´30x	192	193	194	195	196	197	198	→199	″Cx
´31x	₋200	▲201	△202	⫫203	⫩204	▶205	▷206	207	
´32x	208	209	210	▼211	▽212	213	214	◀215	″Dx
´33x	◁216	⟸217	⟹218	⟺219	⟻220	◆221	◇222	⟾223	
´34x	⦿224	◇225	○226	₋227	₌228	◌229	●230	231	″Ex
´35x	232	233	234	235	236	237	238	⇍239	
´36x	⇎240	⇏241	⇐242	⇑243	⇒244	⇓245	⇔246	⇕247	″Fx
´37x	⇖248	⇗249	⇘250	⇙251	⇐252	○253	◇254	⇛255	
	″8	″9	″A	″B	″C	″D	″E	″F	

libertinust1-mathex (largesymbols):

	´0	´1	´2	´3	´4	´5	´6	´7	
´00x	(0	)1	2	3	[4	]5	⟦6	⟧7	″0x
´01x	⌊8	⌋9	⌈10	⌉11	{12	}13	14	15	
´02x	⟨16	⟩17	⟪18	⟫19	20	21	/22	\23	″1x
´03x	(24	)25	26	27	[28	]29	⟦30	⟧31	
´04x	⌊32	⌋33	⌈34	⌉35	{36	}37	38	39	″2x
´05x	⟨40	⟩41	⟪42	⟫43	44	45	/46	\47	
´06x	(48	)49	50	51	[52	]53	⟦54	⟧55	″3x
´07x	⌊56	⌋57	⌈58	⌉59	{60	}61	62	63	
´10x	⟨64	⟩65	⟪66	⟫67	68	69	/70	\71	″4x
´11x	(72	)73	74	75	[76	]77	⟦78	⟧79	
´12x	⌊80	⌋81	⌈82	⌉83	{84	}85	86	87	″5x
´13x	⟨88	⟩89	⟪90	⟫91	92	93	/94	\95	
´14x	⎛96	⎞97	⎡98	⎤99	⎣100	⎦101	⎢102	⎥103	″6x
´15x	⎝104	⎠105	⎧106	⎩107	⎨108	⎬109	⎪110	⎪111	
´16x	⎝112	⎠113	⎢114	⎥115	√116	√117	√118	√119	″7x
´17x	⎢120	⎡121	√122	√123	√124	⎷125	∫126	∫127	

´20x	∬ 128	∭ 129	⨌ 130	⨌ 131	⨍ 132	⨏ 133	⨌ 134	⨌ 135	″8x
´21x	⨍ 136	∫ 137	⨍ 138	⨍ 139	⨍ 140	⨍ 141	⨏ 142	∮ 143	
´22x	⨍ 144	⨌ 145	⨑ 146	⨏ 147	⨑ 148	⨏ 149	150	151	″9x
´23x	152	153	154	155	156	157	158	159	
´26x	176	∏ 177	⨅ 178	∑ 179	⋀ 180	⋁ 181	⋂ 182	⋃ 183	″Bx
´27x	/ 184	\ 185	⨀ 186	⨁ 187	⨂ 188	⨃ 189	⨄ 190	⨅ 191	
´30x	⨆ 192	193	194	195	196	197	198	∏ 199	″Cx
´31x	⨅ 200	∑ 201	⋀ 202	⋁ 203	⋂ 204	⋃ 205	/ 206	\ 207	
´32x	⨀ 208	⨁ 209	⨂ 210	⨃ 211	⨄ 212	⨅ 213	⨆ 214	215	″Dx
´33x	216	217	218	219	(220	) 221	222	223	
´34x	⟦ 224	⟧ 225	⌊ 226	⌋ 227	⌈ 228	⌉ 229	∫ 230	⎫ 231	″Ex
´35x	232	233	⟨ 234	⟩ 235	⟪ 236	⟫ 237	238	239	
´36x	⎪ 240	‖ 241	242	⎪ 243	‖ 244	245	246	247	″Fx
´37x	248	√ 249	250	251	252	253	254	255	
	″8	″9	″A	″B	″C	″D	″E	″F	

libertinust1-mathbb (mathbb):

	´0	´1	´2	´3	´4	´5	´6	´7	
´10x	64	𝔸65	𝔹66	ℂ67	𝔻68	𝔼69	𝔽70	𝔾71	″4x
´11x	ℍ72	𝕀73	𝕁74	𝕂75	𝕃76	𝕄77	ℕ78	𝕆79	
´12x	ℙ80	ℚ81	ℝ82	𝕊83	𝕋84	𝕌85	𝕍86	𝕎87	″5x
´13x	𝕏88	𝕐89	ℤ90	91	92	93	94	95	
´14x	96	𝕒97	𝕓98	𝕔99	𝕕100	𝕖101	𝕗102	𝕘103	″6x
´15x	𝕙104	𝕚105	𝕛106	𝕜107	𝕝108	𝕞109	𝕟110	𝕠111	
´16x	𝕡112	𝕢113	𝕣114	𝕤115	𝕥116	𝕦117	𝕧118	𝕨119	″7x
´17x	𝕩120	𝕪121	𝕫122	𝕝123	𝕁124	125	126	⁀127	
	″8	″9	″A	″B	″C	″D	″E	″F	

New sans serif alphabets:
With the new additions to libertinust1math, starting with version 1.1, there are substantial changes to the handling of sans serif. The changes have the following goals.

- Provide a math package with which all ISO typesetting rules may be followed.
- Simultaneously, add options for the non-ISO user to be able to make better use of sans serif math symbols.
- Provide the option of sans serif Roman and Greek math letters together with math symbols from libertinust1math, which may be useful to those who like to make their slides using sans serif for enhanced readability at lower resolutions.

Thanks go to Claudio Beccari for convincing me that the effort involved was worthwhile, at least for the ISO case.

Four fonts parallel to the serifed roman and italic (regular and bold) have been added, with sans serif alphanumeric symbols based on those in STIX, but slightly reduced in size and the italics reworked to match the italic slant of libertine. (Aside from Biolinum Roman, the Biolinum fonts are not, in my opinion, suitable for use as sans math to accompany libertinust1math.) Glyphs in the four fonts may be accessed by the macros \mathsf, \mathsfbf, \mathsfit and \mathsfbfit, though not all of these may be defined, depending on the options you set.

New Package Options and Macros:
I've followed to some extent the options and macros in the isomath package.

- nosans provides the bare minimum access to sans serif math letters, preserving just \mathsf and its bold counterpart, \mathsfbf. This option keeps the use of additional math families to a minimum. (If you make no use of those macros, no new math families are created.) If nosans is not specified, new symbol fonts sans and sansi are declared, along with bold versions, and \mathsfbfit is declared to be the bold italic sans math alphabet. Two of the available sixteen math families will be committed.
- Option lcgreekalpha allows proper handling of alphabet changes for lower-case Greek letters. (They are mathord by default—this option changes them to mathalpha, so that you do in fact get the expected sans Greek when you enter \mathsf{\phi}.)
- ISO specifies changes needed to conform to ISO typesetting rules. If specified, the following changes will occur.
  - Option nosans is turned off, if present, and option mathsfit is ignored unless the additional option mathsfit is specified.
  - Option lcgreekalpha is turned on, allowing proper handling of alphabet changes for lower-case Greek letters.
  - Option slantedGreek is turned on, as required by ISO rules.
  - New commands are provided: \mathboldsans means \mathsfbfit, and \mathbold means \mathbfit, while each of \vectorsym, \matrixsym means \mathbfit, and \tensorsym means \mathsfbfit. For example, $\matrixsym{A}$ renders as $\boldsymbol{A}$, in accordance with ISO rules.
  - If you add the option mathsfit, then you also have access to sans italic math, which ISO regards as an option for some special cases. This will use another math family.
  - The option reuseMathAlphabets may be added, and can reduce math family usage by redefining \mathbf to mean \mathbfit (math bold italic) and \mathsf to mean \mathsfbfit (sans math bold italic.)
  - As it may be important to keep track of your math families when using ISO, I've added \ShowMathFamilies. If you get an error message "too many math alphabets used in version normal", run the macro just before the error location to get an idea of its cause.
- mathsfit may be used without ISO, with the same effect described under the ISO heading.
- reuseMathAlphabets may be used without ISO, but really makes sense only under ISO.

- `sansmath` sets `nosans` and substitutes sans letters (Roman and Greek) in place of the default Libertine letters. With this option, every math expression in the document will be rendered using sans serif letters.

libertinust1-mathsfrm (\mathsf)

	´0	´1	´2	´3	´4	´5	´6	´7	
´00x	Γ0	Δ1	Θ2	Λ3	Ξ4	Π5	Σ6	Υ7	˝0x
´01x	Φ8	Ψ9	Ω10	α11	β12	γ13	δ14	ϵ15	
´02x	ζ16	η17	θ18	ι19	κ20	λ21	μ22	ν23	˝1x
´03x	ξ24	π25	ρ26	σ27	τ28	υ29	ϕ30	χ31	
´04x	ψ32	ω33	ε34	ϑ35	ϖ36	ϱ37	ς38	φ39	˝2x
´05x	∇40	∂41	−42	+43	±44	∓45	(46	)47	
´06x	048	149	250	351	452	553	654	755	˝3x
´07x	856	957	:58	;59	*60	=61	$62	?63	
´10x	!64	A65	B66	C67	D68	E69	F70	G71	˝4x
´11x	H72	I73	J74	K75	L76	M77	N78	O79	
´12x	P80	Q81	R82	S83	T84	U85	V86	W87	˝5x
´13x	X88	Y89	Z90	[91	\92	]93	{94	/95	
´14x	}96	a97	b98	c99	d100	e101	f102	g103	˝6x
´15x	h104	i105	j106	k107	l108	m109	n110	o111	
´16x	p112	q113	r114	s115	t116	u117	v118	w119	˝7x
´17x	x120	y121	z122	ı123	ȷ124	#125	%126	’127	
´20x	`128	´129	ˆ130	˜131	¯132	˘133	˙134	¨135	˝8x
´21x	˚136	°137	ˇ138	˘139	‘140	’141	¬142	←143	
´22x	→144	←145	→146	...147	148	↔149	⌐150	¬151	˝9x
´23x	*152	&153	@154	¬155	·156	×157	≼158	÷159	
´24x	Ƶ160	⅙161	ɜ162	†163	‡164	•165	..166	...167	˝Ax
´25x	/168	//169	///170	\171	\\172	\\\173	˄174	!!175	
´26x	▪176	⁄177	??178	179	////180	181	182	183	˝Bx
´27x	≍184	185	Ɛ186	ʊ187	ı188	Å189	Ǝ190	Ↄ191	
´30x	⅂192	⅃193	ʎ194	195	⅋196	∀197	Ɔ198	∃199	˝Cx
´31x	ⱯⱯ200	∅201	Δ202	∈203	∉204	ϵ205	ɜ206	ⱯⱯ207	
´32x	ɜ208	▮209	┼210	≥211	\212	∘213	•214	∝215	˝Dx
´33x	∞216	∟217	∠218	∡219	∢220	∣221	∤222	∥223	
´34x	∦224	∧225	∨226	∩227	∪228	∴229	∵230	∅231	˝Ex
´35x	::232	∸233	∹234	∺235	÷236	∼237	∽238	∾239	
´36x	∿240	≀241	≁242	≂243	≃244	≄245	≅246	≇247	˝Fx
´37x	≇248	≈249	≉250	≊251	≋252	≌253	×254	≎255	
	˝8	˝9	˝A	˝B	˝C	˝D	˝E	˝F	

libertinust1-mathsfrm-bold (\mathsfbf)

	´0	´1	´2	´3	´4	´5	´6	´7	
´00x	$\mathbf{\Gamma}_{0}$	$\mathbf{\Delta}_{1}$	$\mathbf{\Theta}_{2}$	$\mathbf{\Lambda}_{3}$	$\mathbf{\Xi}_{4}$	$\mathbf{\Pi}_{5}$	$\mathbf{\Sigma}_{6}$	$\mathbf{\Upsilon}_{7}$	″0x
´01x	$\mathbf{\Phi}_{8}$	$\mathbf{\Psi}_{9}$	$\mathbf{\Omega}_{10}$	$\mathbf{\alpha}_{11}$	$\mathbf{\beta}_{12}$	$\mathbf{\gamma}_{13}$	$\mathbf{\delta}_{14}$	$\mathbf{\epsilon}_{15}$	
´02x	$\mathbf{\zeta}_{16}$	$\mathbf{\eta}_{17}$	$\mathbf{\theta}_{18}$	$\mathbf{\iota}_{19}$	$\mathbf{\kappa}_{20}$	$\mathbf{\lambda}_{21}$	$\mathbf{\mu}_{22}$	$\mathbf{\nu}_{23}$	″1x
´03x	$\mathbf{\xi}_{24}$	$\mathbf{\pi}_{25}$	$\mathbf{\rho}_{26}$	$\mathbf{\sigma}_{27}$	$\mathbf{\tau}_{28}$	$\mathbf{\upsilon}_{29}$	$\mathbf{\phi}_{30}$	$\mathbf{\chi}_{31}$	
´04x	$\mathbf{\psi}_{32}$	$\mathbf{\omega}_{33}$	$\mathbf{\varepsilon}_{34}$	$\mathbf{\vartheta}_{35}$	$\mathbf{\varpi}_{36}$	$\mathbf{\varrho}_{37}$	$\mathbf{\varsigma}_{38}$	$\mathbf{\varphi}_{39}$	″2x
´05x	$\mathbf{\nabla}_{40}$	$\mathbf{\partial}_{41}$	$-_{42}$	$+_{43}$	$\pm_{44}$	$\mp_{45}$	$(_{46}$	$)_{47}$	
´06x	$\mathbf{0}_{48}$	$\mathbf{1}_{49}$	$\mathbf{2}_{50}$	$\mathbf{3}_{51}$	$\mathbf{4}_{52}$	$\mathbf{5}_{53}$	$\mathbf{6}_{54}$	$\mathbf{7}_{55}$	″3x
´07x	$\mathbf{8}_{56}$	$\mathbf{9}_{57}$	$:_{58}$	$;_{59}$	$*_{60}$	$=_{61}$	$\$_{62}$	$\mathbf{?}_{63}$	
´10x	$\mathbf{!}_{64}$	$\mathbf{A}_{65}$	$\mathbf{B}_{66}$	$\mathbf{C}_{67}$	$\mathbf{D}_{68}$	$\mathbf{E}_{69}$	$\mathbf{F}_{70}$	$\mathbf{G}_{71}$	″4x
´11x	$\mathbf{H}_{72}$	$\mathbf{I}_{73}$	$\mathbf{J}_{74}$	$\mathbf{K}_{75}$	$\mathbf{L}_{76}$	$\mathbf{M}_{77}$	$\mathbf{N}_{78}$	$\mathbf{O}_{79}$	
´12x	$\mathbf{P}_{80}$	$\mathbf{Q}_{81}$	$\mathbf{R}_{82}$	$\mathbf{S}_{83}$	$\mathbf{T}_{84}$	$\mathbf{U}_{85}$	$\mathbf{V}_{86}$	$\mathbf{W}_{87}$	″5x
´13x	$\mathbf{X}_{88}$	$\mathbf{Y}_{89}$	$\mathbf{Z}_{90}$	$[_{91}$	$\backslash_{92}$	$]_{93}$	$\{_{94}$	$/_{95}$	
´14x	$\}_{96}$	$\mathbf{a}_{97}$	$\mathbf{b}_{98}$	$\mathbf{c}_{99}$	$\mathbf{d}_{100}$	$\mathbf{e}_{101}$	$\mathbf{f}_{102}$	$\mathbf{g}_{103}$	″6x
´15x	$\mathbf{h}_{104}$	$\mathbf{i}_{105}$	$\mathbf{j}_{106}$	$\mathbf{k}_{107}$	$\mathbf{l}_{108}$	$\mathbf{m}_{109}$	$\mathbf{n}_{110}$	$\mathbf{o}_{111}$	
´16x	$\mathbf{p}_{112}$	$\mathbf{q}_{113}$	$\mathbf{r}_{114}$	$\mathbf{s}_{115}$	$\mathbf{t}_{116}$	$\mathbf{u}_{117}$	$\mathbf{v}_{118}$	$\mathbf{w}_{119}$	″7x
´17x	$\mathbf{x}_{120}$	$\mathbf{y}_{121}$	$\mathbf{z}_{122}$	$\mathbf{\imath}_{123}$	$\mathbf{\jmath}_{124}$	$\#_{125}$	$\%_{126}$	$'_{127}$	
´20x	$\grave{}_{128}$	$\acute{}_{129}$	$\hat{}_{130}$	$\tilde{}_{131}$	$\bar{}_{132}$	$\breve{}_{133}$	$\dot{}_{134}$	$\ddot{}_{135}$	″8x
´21x	$\mathring{}_{136}$	$\circ_{137}$	$\check{}_{138}$	$\mathring{}_{139}$	$\grave{}_{140}$	$\acute{}_{141}$	$\urcorner_{142}$	$\leftarrow_{143}$	
´22x	$\rightarrow_{144}$	$\leftarrow_{145}$	$\rightarrow_{146}$	$\cdots_{147}$	$\cdots_{148}$	$\leftrightarrow_{149}$	$\top_{150}$	$\neg_{151}$	″9x
´23x	$*_{152}$	$\&_{153}$	$@_{154}$	$\neg_{155}$	$\cdot_{156}$	$\times_{157}$	$\preceq_{158}$	$\div_{159}$	
´24x	$\mathbb{Z}_{160}$	$\nicefrac{1}{6}_{161}$	$\ni_{162}$	$\dagger_{163}$	$\ddagger_{164}$	$\bullet_{165}$	$..._{166}$	$\ldots_{167}$	″Ax
´25x	$/_{168}$	$//_{169}$	$///_{170}$	$\backslash_{171}$	$\backslash\backslash_{172}$	$\backslash\backslash\backslash_{173}$	$\wedge_{174}$	$!!_{175}$	
´26x	$\blacksquare_{176}$	$/_{177}$	$\mathbf{??}_{178}$	$_{179}$	$////_{180}$	$\nmid_{181}$	$_{182}$	$_{183}$	″Bx
´27x	$\asymp_{184}$	$_{185}$	$\mathfrak{E}_{186}$	$\mathfrak{O}_{187}$	$\imath_{188}$	$\mathring{A}_{189}$	$\exists_{190}$	$\Im_{191}$	
´30x	$\urcorner_{192}$	$\bot_{193}$	λ_{194}	$_{195}$	$\mathfrak{B}_{196}$	$\forall_{197}$	$\complement_{198}$	$\exists_{199}$	″Cx
´31x	$\nexists_{200}$	$\emptyset_{201}$	$\mathbf{\Delta}_{202}$	$\in_{203}$	$\notin_{204}$	$\in_{205}$	$\ni_{206}$	$\not\ni_{207}$	
´32x	$\ni_{208}$	$\blacksquare_{209}$	$\dotplus_{210}$	$\succeq_{211}$	$\backslash_{212}$	$\circ_{213}$	$\bullet_{214}$	$\propto_{215}$	″Dx
´33x	∞_{216}	$\llcorner_{217}$	$\angle_{218}$	$\measuredangle_{219}$	$\sphericalangle_{220}$	$\mid_{221}$	$\nmid_{222}$	$\parallel_{223}$	
´34x	$\nparallel_{224}$	$\wedge_{225}$	$\vee_{226}$	$\cap_{227}$	$\cup_{228}$	$\therefore_{229}$	$\because_{230}$	$\emptyset_{231}$	″Ex
´35x	$::_{232}$	$\dot{-}_{233}$	$-:_{234}$	$\dot{\div}_{235}$	$\div_{236}$	$\sim_{237}$	$\backsim_{238}$	$\wr_{239}$	
´36x	$\backsim_{240}$	$\backsim_{241}$	$\eqslantless_{242}$	$\simeq_{243}$	$\simeq_{244}$	$\not\simeq_{245}$	$\cong_{246}$	$\not\cong_{247}$	″Fx
´37x	$\not\cong_{248}$	$\approx_{249}$	$\not\approx_{250}$	$\approx_{251}$	$\approx_{252}$	$\eqsim_{253}$	$\bowtie_{254}$	$\fallingdotseq_{255}$	
	″8	″9	″A	″B	″C	″D	″E	″F	

libertinust1-mathsfit (\mathsfit)

	´0	´1	´2	´3	´4	´5	´6	´7	
´00x	Γ_{0}	Δ_{1}	Θ_{2}	Λ_{3}	Ξ_{4}	Π_{5}	Σ_{6}	Υ_{7}	˝0x
´01x	Φ_{8}	Ψ_{9}	Ω_{10}	α_{11}	β_{12}	γ_{13}	δ_{14}	ϵ_{15}	
´02x	ζ_{16}	η_{17}	θ_{18}	ι_{19}	κ_{20}	λ_{21}	μ_{22}	ν_{23}	˝1x
´03x	ξ_{24}	π_{25}	ρ_{26}	σ_{27}	τ_{28}	υ_{29}	ϕ_{30}	χ_{31}	
´04x	ψ_{32}	ω_{33}	ε_{34}	ϑ_{35}	ϖ_{36}	ϱ_{37}	ς_{38}	φ_{39}	˝2x
´05x	∇_{40}	∂_{41}	$\aleph_{42}$	$\beth_{43}$	$\gimel_{44}$	$\daleth_{45}$	$\triangleright_{46}$	$\triangleleft_{47}$	
´06x	0_{48}	1_{49}	2_{50}	3_{51}	4_{52}	5_{53}	6_{54}	7_{55}	˝3x
´07x	8_{56}	9_{57}	$.{}_{58}$	$,{}_{59}$	$<_{60}$	$\hbar_{61}$	$>_{62}$	$\star_{63}$	
´10x	$\lneqq_{64}$	A_{65}	B_{66}	C_{67}	D_{68}	E_{69}	F_{70}	G_{71}	˝4x
´11x	H_{72}	I_{73}	J_{74}	K_{75}	L_{76}	M_{77}	N_{78}	O_{79}	
´12x	P_{80}	Q_{81}	R_{82}	S_{83}	T_{84}	U_{85}	V_{86}	W_{87}	˝5x
´13x	X_{88}	Y_{89}	Z_{90}	$\flat_{91}$	$\natural_{92}$	$\sharp_{93}$	$\smile_{94}$	$\frown_{95}$	
´14x	$\hbar_{96}$	a_{97}	b_{98}	c_{99}	d_{100}	e_{101}	f_{102}	g_{103}	˝6x
´15x	h_{104}	i_{105}	j_{106}	k_{107}	l_{108}	m_{109}	n_{110}	o_{111}	
´16x	p_{112}	q_{113}	r_{114}	s_{115}	t_{116}	u_{117}	v_{118}	w_{119}	˝7x
´17x	x_{120}	y_{121}	z_{122}	$\imath_{123}$	$\jmath_{124}$	$\gneqq_{125}$	$\ll_{126}$	$\frown_{127}$	
´20x	$\grave{}_{128}$	$\acute{}_{129}$	$\hat{}_{130}$	$\tilde{}_{131}$	$\bar{}_{132}$	$\breve{}_{133}$	$\dot{}_{134}$	$\ddot{}_{135}$	˝8x
´21x	$\mathring{}_{136}$	$\mathring{}_{137}$	$\check{}_{138}$	$\breve{}_{139}$	$\grave{}_{140}$	$\acute{}_{141}$	$\ddot{}_{142}$	$\tilde{}_{143}$	
´22x	$\vec{}_{144}$	$\overleftarrow{}_{145}$	$\vec{}_{146}$	$\dddot{}_{147}$	$\ddddot{}_{148}$	$\overleftrightarrow{}_{149}$	$\overline{}_{150}$	$\overline{}_{151}$	˝9x
´23x	$*_{152}$	$\bar{}_{153}$	$\widehat{}_{154}$	$\tilde{}_{155}$	$\breve{}_{156}$	$\widehat{}_{157}$	$\tilde{}_{158}$	$\breve{}_{159}$	
´24x	$\widehat{}_{160}$	$\tilde{}_{161}$	$\breve{}_{162}$	$\widehat{}_{163}$	$\tilde{}_{164}$	$\breve{}_{165}$	$\widehat{}_{166}$	$\tilde{}_{167}$	˝Ax
´25x	$\breve{}_{168}$	$\frown_{169}$	$\neg_{170}$	$\backsim_{171}$	$\backsim_{172}$	$\backsim_{173}$	$\backsim_{174}$	$\backsim_{175}$	
´26x	$\overline{}_{176}$	$\overline{}_{177}$	$\underline{}_{178}$	$\underline{}_{179}$	$\ulcorner_{180}$	$\urcorner_{181}$	$\llcorner_{182}$	$\lrcorner_{183}$	˝Bx
´27x	$\gg_{184}$	$\ll_{185}$	$\not\prec_{186}$	$\not\succ_{187}$	$\succ_{188}$	$\not\preceq_{189}$	$\not\succeq_{190}$	$\lesssim_{191}$	
´30x	$\gtrsim_{192}$	$\not\lesssim_{193}$	$\not\gtrsim_{194}$	$\lesssim_{195}$	$\gtrsim_{196}$	$\not\lesssim_{197}$	$\not\gtrsim_{198}$	$\prec_{199}$	˝Cx
´31x	$\succ_{200}$	$\preccurlyeq_{201}$	$\succcurlyeq_{202}$	$\precsim_{203}$	$\succsim_{204}$	$\not\prec_{205}$	$\not\succ_{206}$	$\subset_{207}$	
´32x	$\supset_{208}$	$\not\subset_{209}$	$\not\supset_{210}$	$\subseteq_{211}$	$\supseteq_{212}$	$\not\subseteq_{213}$	$\not\supseteq_{214}$	$\subsetneq_{215}$	˝Dx
´33x	$\supsetneq_{216}$	$\uplus_{217}$	$\uplus_{218}$	$\uplus_{219}$	$\sqsubset_{220}$	$\sqsupset_{221}$	$\sqsubseteq_{222}$	$\sqsupseteq_{223}$	
´34x	$\sqcap_{224}$	$\sqcup_{225}$	$\oplus_{226}$	$\ominus_{227}$	$\otimes_{228}$	$\oslash_{229}$	$\odot_{230}$	$\circledcirc_{231}$	˝Ex
´35x	$\circledast_{232}$	$\circleddash_{233}$	$\ominus_{234}$	$\boxplus_{235}$	$\boxminus_{236}$	$\boxtimes_{237}$	$\boxdot_{238}$	$\vdash_{239}$	
´36x	$\dashv_{240}$	$\top_{241}$	$\bot_{242}$	$\vdash_{243}$	$\models_{244}$	$\models_{245}$	$\Vdash_{246}$	$\Vdash_{247}$	˝Fx
´37x	$\models_{248}$	$\nvdash_{249}$	$\nvDash_{250}$	$\nVdash_{251}$	$\nVDash_{252}$	$_{253}$	$\boldsymbol{x}_{254}$	$\trianglelefteq_{255}$	
	˝8	˝9	˝A	˝B	˝C	˝D	˝E	˝F	

libertinust1-mathsfit-bold (\mathsfbfit)

	´0	´1	´2	´3	´4	´5	´6	´7	
´00x	$\mathit{\Gamma}_{0}$	$\mathit{\Delta}_{1}$	$\mathit{\Theta}_{2}$	$\mathit{\Lambda}_{3}$	$\mathit{\Xi}_{4}$	$\mathit{\Pi}_{5}$	$\mathit{\Sigma}_{6}$	$\mathit{\Upsilon}_{7}$	˝0x
´01x	$\mathit{\Phi}_{8}$	$\mathit{\Psi}_{9}$	$\mathit{\Omega}_{10}$	$\mathit{\alpha}_{11}$	$\mathit{\beta}_{12}$	$\mathit{\gamma}_{13}$	$\mathit{\delta}_{14}$	$\mathit{\epsilon}_{15}$	
´02x	$\mathit{\zeta}_{16}$	$\mathit{\eta}_{17}$	$\mathit{\theta}_{18}$	$\mathit{\iota}_{19}$	$\mathit{\kappa}_{20}$	$\mathit{\lambda}_{21}$	$\mathit{\mu}_{22}$	$\mathit{\nu}_{23}$	˝1x
´03x	$\mathit{\xi}_{24}$	$\mathit{\pi}_{25}$	$\mathit{\rho}_{26}$	$\mathit{\sigma}_{27}$	$\mathit{\tau}_{28}$	$\mathit{\upsilon}_{29}$	$\mathit{\phi}_{30}$	$\mathit{\chi}_{31}$	
´04x	$\mathit{\psi}_{32}$	$\mathit{\omega}_{33}$	$\mathit{\varepsilon}_{34}$	$\mathit{\vartheta}_{35}$	$\mathit{\varpi}_{36}$	$\mathit{\varrho}_{37}$	$\mathit{\varsigma}_{38}$	$\mathit{\varphi}_{39}$	˝2x
´05x	∇_{40}	∂_{41}	$\aleph_{42}$	$\beth_{43}$	$\gimel_{44}$	$\daleth_{45}$	$\triangleright_{46}$	$\triangleleft_{47}$	
´06x	$\mathbf{0}_{48}$	$\mathbf{1}_{49}$	$\mathbf{2}_{50}$	$\mathbf{3}_{51}$	$\mathbf{4}_{52}$	$\mathbf{5}_{53}$	$\mathbf{6}_{54}$	$\mathbf{7}_{55}$	˝3x
´07x	$\mathbf{8}_{56}$	$\mathbf{9}_{57}$	.58	,59	$<_{60}$	$\mathit{\hbar}_{61}$	$>_{62}$	$\star_{63}$	
´10x	$\lneqq_{64}$	A_{65}	B_{66}	C_{67}	D_{68}	E_{69}	F_{70}	G_{71}	˝4x
´11x	H_{72}	I_{73}	J_{74}	K_{75}	L_{76}	M_{77}	N_{78}	O_{79}	
´12x	P_{80}	Q_{81}	R_{82}	S_{83}	T_{84}	U_{85}	V_{86}	W_{87}	˝5x
´13x	X_{88}	Y_{89}	Z_{90}	$\flat_{91}$	$\natural_{92}$	$\sharp_{93}$	$\smile_{94}$	$\frown_{95}$	
´14x	$\mathit{\hbar}_{96}$	a_{97}	b_{98}	c_{99}	d_{100}	e_{101}	f_{102}	g_{103}	˝6x
´15x	h_{104}	i_{105}	j_{106}	k_{107}	l_{108}	m_{109}	n_{110}	o_{111}	
´16x	p_{112}	q_{113}	r_{114}	s_{115}	t_{116}	u_{117}	v_{118}	w_{119}	˝7x
´17x	x_{120}	y_{121}	z_{122}	$\imath_{123}$	$\jmath_{124}$	$\gneqq_{125}$	$\ll_{126}$	$\frown_{127}$	
´20x	ˋ128	´129	ˆ130	˜131	¯132	˘133	˙134	¨135	˝8x
´21x	˚136	˝137	ˇ138	˘139	˛140	¸141	˛142	˝143	
´22x	$\rightharpoonup_{144}$	$\leftarrow_{145}$	$\rightarrow_{146}$	...147	148	$\leftrightarrow_{149}$	¬150	⌐151	˝9x
´23x	*152	−153	ˆ154	˜155	ˇ156	ˆ157	˜158	ˇ159	
´24x	‿160	⁀161	˘162	‾163	⁀164	‿165	‾166	⁀167	˝Ax
´25x	‿168	⁀169	¬170	⌣171	⁀172	‿173	⁀174	⁀175	
´26x	⁀176	‿177	‿178	⁀179	⌐180	¬181	⌐182	⌐183	˝Bx
´27x	$\gg_{184}$	⟩185	$\not\prec_{186}$	$\not<_{187}$	$\not>_{188}$	$\not\leq_{189}$	$\not\geq_{190}$	$\lesssim_{191}$	
´30x	$\gtrsim_{192}$	$\not\lesssim_{193}$	$\not\gtrsim_{194}$	$\lesssim_{195}$	$\gtrsim_{196}$	$\not\lesssim_{197}$	$\not\gtrsim_{198}$	$\prec_{199}$	˝Cx
´31x	$\succ_{200}$	$\preccurlyeq_{201}$	$\succcurlyeq_{202}$	$\precsim_{203}$	$\succsim_{204}$	$\not\prec_{205}$	$\not\succ_{206}$	$\subset_{207}$	
´32x	$\supset_{208}$	$\not\subset_{209}$	$\not\supset_{210}$	$\subseteq_{211}$	$\supseteq_{212}$	$\not\subseteq_{213}$	$\not\supseteq_{214}$	$\subsetneq_{215}$	˝Dx
´33x	$\supsetneq_{216}$	$\uplus_{217}$	$\uplus_{218}$	$\uplus_{219}$	$\sqsubset_{220}$	$\sqsupset_{221}$	$\sqsubseteq_{222}$	$\sqsupseteq_{223}$	
´34x	$\sqcap_{224}$	$\sqcup_{225}$	$\oplus_{226}$	$\ominus_{227}$	$\otimes_{228}$	$\oslash_{229}$	$\odot_{230}$	$\circledcirc_{231}$	˝Ex
´35x	$\circledast_{232}$	$\circleddash_{233}$	$\ominus_{234}$	$\boxplus_{235}$	$\boxminus_{236}$	$\boxtimes_{237}$	$\boxdot_{238}$	$\vdash_{239}$	
´36x	$\dashv_{240}$	$\top_{241}$	$\bot_{242}$	$\vdash_{243}$	$\models_{244}$	$\models_{245}$	$\Vdash_{246}$	$\Vdash_{247}$	˝Fx
´37x	$\models_{248}$	$\nvdash_{249}$	$\nvDash_{250}$	$\nVdash_{251}$	$\nVDash_{252}$	253	x_{254}	$\trianglelefteq_{255}$	
	˝8	˝9	˝A	˝B	˝C	˝D	˝E	˝F	

Brief samples using the sans serif options, showing mathgroup usage

1.

`Options=[]`, and no use of `\mathsfxxxx` macros.

$y = f(x)$.

```
*** Mathgroups ***
(0: \LS1/libertinust1math/m/n/12 = libertinust1-mathrm at 14.39996pt [operators])
(1: \LS1/libertinust1math/m/it/12 = libertinust1-mathit at 14.39996pt [letters])
(2: \LS2/libertinust1mathsym/m/n/12 = libertinust1-mathsym at 14.39996pt [symbols])
(3: \LS2/libertinust1mathex/m/n/12 = libertinust1-mathex at 14.39996pt [largesymbols])
(4: \LS1/libertinust1mathbb/m/n/12 = libertinust1-mathbb at 14.39996pt [symbolsbb])
(5: \LS1/libertinust1math/b/n/12 = libertinust1-mathrm-bold at 14.39996pt [bold-operators])
(6: \LS1/libertinust1math/b/it/12 = libertinust1-mathit-bold at 14.39996pt [bold-letters])
```

2.

`Options=[]`

macro	symbol
`\mathsf{x}`	x
`\mathsfbf{y}`	**y**
`\mathsfbfit{z}`	***z***
`\mathsfit{w}`	undefined

```
*** Mathgroups ***
(0: \LS1/libertinust1math/m/n/12 = libertinust1-mathrm at 14.39996pt [operators])
(1: \LS1/libertinust1math/m/it/12 = libertinust1-mathit at 14.39996pt [letters])
(2: \LS2/libertinust1mathsym/m/n/12 = libertinust1-mathsym at 14.39996pt [symbols])
(3: \LS2/libertinust1mathex/m/n/12 = libertinust1-mathex at 14.39996pt [largesymbols])
(4: \LS1/libertinust1mathbb/m/n/12 = libertinust1-mathbb at 14.39996pt [symbolsbb])
(5: \LS1/libertinust1math/b/n/12 = libertinust1-mathrm-bold at 14.39996pt [bold-operators])
(6: \LS1/libertinust1math/b/it/12 = libertinust1-mathit-bold at 14.39996pt [bold-letters])
(7: \LS1/libertinust1mathsf/m/n/12 = libertinust1-mathsfrm at 14.39996pt)
(8: \LS1/libertinust1mathsf/b/n/12 = libertinust1-mathsfrm-bold at 14.39996pt)
(9: \LS1/libertinust1mathsf/b/it/12 = libertinust1-mathsfit-bold at 14.39996pt)
```

3.

`Options=[nosans]`, and no use of `\mathsfxxxx` macros.

$y = f(x)$.

```
*** Mathgroups ***
(0: \LS1/libertinust1math/m/n/12 = libertinust1-mathrm at 14.39996pt [operators])
(1: \LS1/libertinust1math/m/it/12 = libertinust1-mathit at 14.39996pt [letters])
(2: \LS2/libertinust1mathsym/m/n/12 = libertinust1-mathsym at 14.39996pt [symbols])
(3: \LS2/libertinust1mathex/m/n/12 = libertinust1-mathex at 14.39996pt [largesymbols])
```

```
(4: \LS1/libertinust1mathbb/m/n/12 = libertinust1-mathbb at 14.39996pt [symbolsbb])
(5: \LS1/libertinust1math/b/n/12 = libertinust1-mathrm-bold at 14.39996pt [bold-operators])
(6: \LS1/libertinust1math/b/it/12 = libertinust1-mathit-bold at 14.39996pt [bold-letters])
```

4.

`Options=[nosans]`, and no use of `\mathsfxxxx` macros

macro	symbol
`\mathsf{x}`	**x**
`\mathsfbf{y}`	**y**
`\mathsfbfit{z}`	undefined
`\mathsfit{w}`	undefined

```
*** Mathgroups ***
(0: \LS1/libertinust1math/m/n/12 = libertinust1-mathrm at 14.39996pt [operators])
(1: \LS1/libertinust1math/m/it/12 = libertinust1-mathit at 14.39996pt [letters])
(2: \LS2/libertinust1mathsym/m/n/12 = libertinust1-mathsym at 14.39996pt [symbols])
(3: \LS2/libertinust1mathex/m/n/12 = libertinust1-mathex at 14.39996pt [largesymbols])
(4: \LS1/libertinust1mathbb/m/n/12 = libertinust1-mathbb at 14.39996pt [symbolsbb])
(5: \LS1/libertinust1math/b/n/12 = libertinust1-mathrm-bold at 14.39996pt [bold-operators])
(6: \LS1/libertinust1math/b/it/12 = libertinust1-mathit-bold at 14.39996pt [bold-letters])
(7: \LS1/libertinust1mathsf/m/n/12 = libertinust1-mathsfrm at 14.39996pt)
(8: \LS1/libertinust1mathsf/b/n/12 = libertinust1-mathsfrm-bold at 14.39996pt)
```

5.

`Options=[mathsfit]`, and no use of `\mathsfxxxx` macros.

$y = f(x)$.

```
*** Mathgroups ***
(0: \LS1/libertinust1math/m/n/12 = libertinust1-mathrm at 14.39996pt [operators])
(1: \LS1/libertinust1math/m/it/12 = libertinust1-mathit at 14.39996pt [letters])
(2: \LS2/libertinust1mathsym/m/n/12 = libertinust1-mathsym at 14.39996pt [symbols])
(3: \LS2/libertinust1mathex/m/n/12 = libertinust1-mathex at 14.39996pt [largesymbols])
(4: \LS1/libertinust1mathbb/m/n/12 = libertinust1-mathbb at 14.39996pt [symbolsbb])
(5: \LS1/libertinust1math/b/n/12 = libertinust1-mathrm-bold at 14.39996pt [bold-operators])
(6: \LS1/libertinust1math/b/it/12 = libertinust1-mathit-bold at 14.39996pt [bold-letters])
```

6.

`Options=[mathsfit]`

macro	symbol
`\mathsf{x}`	**x**
`\mathsfbf{y}`	**y**
`\mathsfbfit{z}`	***z***
`\mathsfit{w}`	***w***

```
*** Mathgroups ***
(0: \LS1/libertinust1math/m/n/12 = libertinust1-mathrm at 14.39996pt [operators])
(1: \LS1/libertinust1math/m/it/12 = libertinust1-mathit at 14.39996pt [letters])
(2: \LS2/libertinust1mathsym/m/n/12 = libertinust1-mathsym at 14.39996pt [symbols])
(3: \LS2/libertinust1mathex/m/n/12 = libertinust1-mathex at 14.39996pt [largesymbols])
(4: \LS1/libertinust1mathbb/m/n/12 = libertinust1-mathbb at 14.39996pt [symbolsbb])
(5: \LS1/libertinust1math/b/n/12 = libertinust1-mathrm-bold at 14.39996pt [bold-operators])
(6: \LS1/libertinust1math/b/it/12 = libertinust1-mathit-bold at 14.39996pt [bold-letters])
(7: \LS1/libertinust1mathsf/m/n/12 = libertinust1-mathsfrm at 14.39996pt)
(8: \LS1/libertinust1mathsf/b/n/12 = libertinust1-mathsfrm-bold at 14.39996pt)
(9: \LS1/libertinust1mathsf/b/it/12 = libertinust1-mathsfit-bold at 14.39996pt)
(10: \LS1/libertinust1mathsf/m/it/12 = libertinust1-mathsfit at 14.39996pt)
```

7.

`Options=[reuseMathAlphabets]`

macro	symbol
`\mathsf{x}`	x
`\mathsfbf{y}`	$\mathbf{y}$
`\mathsfbfit{z}`	$\boldsymbol{z}$
`\mathsfit{w}`	undefined

```
*** Mathgroups ***
(0: \LS1/libertinust1math/m/n/12 = libertinust1-mathrm at 14.39996pt [operators])
(1: \LS1/libertinust1math/m/it/12 = libertinust1-mathit at 14.39996pt [letters])
(2: \LS2/libertinust1mathsym/m/n/12 = libertinust1-mathsym at 14.39996pt [symbols])
(3: \LS2/libertinust1mathex/m/n/12 = libertinust1-mathex at 14.39996pt [largesymbols])
(4: \LS1/libertinust1mathbb/m/n/12 = libertinust1-mathbb at 14.39996pt [symbolsbb])
(5: \LS1/libertinust1math/b/n/12 = libertinust1-mathrm-bold at 14.39996pt [bold-operators])
(6: \LS1/libertinust1math/b/it/12 = libertinust1-mathit-bold at 14.39996pt [bold-letters])
(7: \LS1/libertinust1mathsf/m/n/12 = libertinust1-mathsfrm at 14.39996pt)
(8: \LS1/libertinust1mathsf/b/n/12 = libertinust1-mathsfrm-bold at 14.39996pt)
(9: \LS1/libertinust1mathsf/b/it/12 = libertinust1-mathsfit-bold at 14.39996pt)
```

8.

`Options=[ISO]`

macro	symbol
`\mathsf{x}`	x
`\mathsfbf{y}`	$\mathbf{y}$
`\mathsfbfit{z}`	$\boldsymbol{z}$
`\mathsfit{w}`	undefined

```
*** Mathgroups ***
(0: \LS1/libertinust1math/m/n/12 = libertinust1-mathrm at 14.39996pt [operators])
(1: \LS1/libertinust1math/m/it/12 = libertinust1-mathit at 14.39996pt [letters])
(2: \LS2/libertinust1mathsym/m/n/12 = libertinust1-mathsym at 14.39996pt [symbols])
```

```
(3: \LS2/libertinust1mathex/m/n/12 = libertinust1-mathex at 14.39996pt [largesymbols])
(4: \LS1/libertinust1mathbb/m/n/12 = libertinust1-mathbb at 14.39996pt [symbolsbb])
(5: \LS1/libertinust1math/b/n/12 = libertinust1-mathrm-bold at 14.39996pt [bold-operators])
(6: \LS1/libertinust1math/b/it/12 = libertinust1-mathit-bold at 14.39996pt [bold-letters])
(7: \LS1/libertinust1mathsf/m/n/12 = libertinust1-mathsfrm at 14.39996pt)
(8: \LS1/libertinust1mathsf/b/n/12 = libertinust1-mathsfrm-bold at 14.39996pt)
(9: \LS1/libertinust1mathsf/b/it/12 = libertinust1-mathsfit-bold at 14.39996pt)
```

9.

`Options=[ISO,mathsfit]`

macro	symbol
`\mathsf{x}`	x
`\mathsfbf{y}`	$\mathbf{y}$
`\mathsfbfit{z}`	$\boldsymbol{z}$
`\mathsfit{w}`	w

```
*** Mathgroups ***
(0: \LS1/libertinust1math/m/n/12 = libertinust1-mathrm at 14.39996pt [operators])
(1: \LS1/libertinust1math/m/it/12 = libertinust1-mathit at 14.39996pt [letters])
(2: \LS2/libertinust1mathsym/m/n/12 = libertinust1-mathsym at 14.39996pt [symbols])
(3: \LS2/libertinust1mathex/m/n/12 = libertinust1-mathex at 14.39996pt [largesymbols])
(4: \LS1/libertinust1mathbb/m/n/12 = libertinust1-mathbb at 14.39996pt [symbolsbb])
(5: \LS1/libertinust1math/b/n/12 = libertinust1-mathrm-bold at 14.39996pt [bold-operators])
(6: \LS1/libertinust1math/b/it/12 = libertinust1-mathit-bold at 14.39996pt [bold-letters])
(7: \LS1/libertinust1mathsf/m/n/12 = libertinust1-mathsfrm at 14.39996pt)
(8: \LS1/libertinust1mathsf/b/n/12 = libertinust1-mathsfrm-bold at 14.39996pt)
(9: \LS1/libertinust1mathsf/b/it/12 = libertinust1-mathsfit-bold at 14.39996pt)
(10: \LS1/libertinust1mathsf/m/it/12 = libertinust1-mathsfit at 14.39996pt)
```

10.

`Options=[ISO,mathsfit,reuseMathAlphabets]`

macro	symbol
`\mathbf{y}`	y
`\mathsf{x}`	$\boldsymbol{x}$
`\mathsfbf{y}`	$\mathbf{y}$
`\mathsfbfit{z}`	$\boldsymbol{z}$
`\mathsfit{w}`	w

```
*** Mathgroups ***
(0: \LS1/libertinust1math/m/n/12 = libertinust1-mathrm at 14.39996pt [operators])
(1: \LS1/libertinust1math/m/it/12 = libertinust1-mathit at 14.39996pt [letters])
(2: \LS2/libertinust1mathsym/m/n/12 = libertinust1-mathsym at 14.39996pt [symbols])
(3: \LS2/libertinust1mathex/m/n/12 = libertinust1-mathex at 14.39996pt [largesymbols])
(4: \LS1/libertinust1mathbb/m/n/12 = libertinust1-mathbb at 14.39996pt [symbolsbb])
(5: \LS1/libertinust1math/b/n/12 = libertinust1-mathrm-bold at 14.39996pt [bold-operators])
```

```
(6: \LS1/libertinust1math/b/it/12 = libertinust1-mathit-bold at 14.39996pt [bold-letters])
(7: \LS1/libertinust1mathsf/b/n/12 = libertinust1-mathsfrm-bold at 14.39996pt)
(8: \LS1/libertinust1mathsf/b/it/12 = libertinust1-mathsfit-bold at 14.39996pt)
(9: \LS1/libertinust1mathsf/m/it/12 = libertinust1-mathsfit at 14.39996pt)
```

11: An example of sansmath.

```
Options=[sansmath]
```

The boundedness of Φ_0 then yields

$$\int_D |\overline{\partial} u|^2 e^{\alpha|z|^2} \geq c_6 \alpha \int_D |u|^2 e^{\alpha|z|^2} + c_7 \delta^{-2} \int_A |u|^2 e^{\alpha|z|^2}.$$

Let $B(X)$ be the set of blocks of Λ_X and let $b(X) := |B(X)|$. If $\varphi \in Q_X$ then φ is constant on the blocks of Λ_X.

$$P_X = \{\varphi \in M \mid \Lambda_\varphi = \Lambda_X\}, \qquad Q_X = \{\varphi \in M \mid \Lambda_\varphi \geq \Lambda_X\}. \tag{1}$$

If $\Lambda_\varphi \geq \Lambda_X$ then $\Lambda_\varphi = \Lambda_Y$ for some $Y \geq X$ so that

$$Q_X = \bigcup_{Y \geq X} P_Y.$$

Thus by Möbius inversion

$$P_Y = \sum_{X \geq Y} \mu(Y, X) Q_{\hat{X}}.$$

Thus there is a bijection from Q_X to $W^{B(X)}$. In particular $|Q_X| = w^{b(X)}$.

```
*** Mathgroups ***
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(1: \LS1/libertinust1math/m/it/12 = libertinust1-mathsfit at 14.39996pt [letters])
(2: \LS2/libertinust1mathsym/m/n/12 = libertinust1-mathsym at 14.39996pt [symbols])
(3: \LS2/libertinust1mathex/m/n/12 = libertinust1-mathex at 14.39996pt [largesymbols])
(4: \LS1/libertinust1mathbb/m/n/12 = libertinust1-mathbb at 14.39996pt [symbolsbb])
(5: \LS1/libertinust1math/b/n/12 = libertinust1-mathsfrm-bold at 14.39996pt [bold-operators])
(6: \LS1/libertinust1math/b/it/12 = libertinust1-mathsfit-bold at 14.39996pt [bold-letters])
```